

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 14: Equation of a Line

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Forms of the equation

1. $ABCD$ is a parallelogram with vertices $A(3, 3)$, $B(0, -2)$, $C(-4, -2)$ and $D(-1, 3)$, read from a graph. (a) Find the point P where the diagonals meet. (b) Find the slopes of CB and DA and check that these sides are parallel. (c) Find the equation of the diagonal AC . **ICSE 2025, 5 marks**

SOLUTION

(a) The diagonals of a parallelogram bisect each other, so P is the mid-point of AC : $P = \left(\frac{3 + (-4)}{2}, \frac{3 + (-2)}{2} \right) = \left(-\frac{1}{2}, \frac{1}{2} \right)$.

(b) Slope of $CB = \frac{-2 - (-2)}{0 - (-4)} = 0$ and slope of $DA = \frac{3 - 3}{-1 - (-4)} = 0$. The slopes are equal, so $CB \parallel DA$.

(c) Slope of $AC = \frac{3 - (-2)}{3 - (-4)} = \frac{5}{7}$. Using $A(3, 3)$: $y - 3 = \frac{5}{7}(x - 3)$, so $7y - 21 = 5x - 15$.

Answer: (a) $P\left(-\frac{1}{2}, \frac{1}{2}\right)$ **(b)** both slopes are 0 **(c)** $5x - 7y + 6 = 0$

2. Triangle ABC has vertices $A(1, -1)$, $B(0, 4)$ and $C(-6, 4)$, and D is the mid-point of BC . Find (a) the coordinates of D (b) the equation of the median AD . **ICSE 2023, 3 marks**

SOLUTION

(a) $D = \left(\frac{0 + (-6)}{2}, \frac{4 + 4}{2} \right) = (-3, 4)$

(b) Slope of $AD = \frac{4 - (-1)}{-3 - 1} = -\frac{5}{4}$. Through A : $y + 1 = -\frac{5}{4}(x - 1)$, so $4y + 4 = -5x + 5$.

Answer: (a) $D(-3, 4)$ **(b)** $5x + 4y - 1 = 0$

2 Lines parallel or perpendicular to a given line

3. Which line passes through the origin and is parallel to $3x + 4y + 7 = 0$? (a) $3x + 4y + 5 = 0$ (b) $4x - 3y - 5 = 0$ (c) $4x - 3y = 0$ (d) $3x + 4y = 0$ **ICSE 2025, 1 mark**

SOLUTION

A parallel line has the same x and y coefficients: $3x + 4y + k = 0$.

It passes through $(0, 0)$, so $k = 0$.

Answer: (d)

4. The line PQ meets the positive x -axis at P , 5 units from the origin, and the negative y -axis at Q , 3 units from the origin. The line RS passes through the origin and is perpendicular to PQ . Find (a) the coordinates of P and Q (b) the equation of RS . **ICSE 2025, 3 marks**

SOLUTION

(a) $P(5, 0)$ and $Q(0, -3)$.

(b) Slope of $PQ = \frac{0 - (-3)}{5 - 0} = \frac{3}{5}$, so the slope of RS is $-\frac{5}{3}$.

RS passes through the origin: $y = -\frac{5}{3}x$.

Answer: (a) $P(5, 0)$, $Q(0, -3)$ (b) $5x + 3y = 0$

5. (a) The lines $kx - y + 4 = 0$ and $2y = 6x + 7$ are perpendicular. Find k . (b) Find the equation of the line parallel to $2y = 6x + 7$ that passes through $(-1, 1)$. **ICSE 2024, 3 marks**

SOLUTION

(a) The first line is $y = kx + 4$, with slope k . The second is $y = 3x + \frac{7}{2}$, with slope 3.

Perpendicular lines: $k \times 3 = -1$, so $k = -\frac{1}{3}$.

(b) A parallel line has slope 3: $y - 1 = 3(x + 1)$

Answer: (a) $k = -\frac{1}{3}$ (b) $y = 3x + 4$

6. The lines $5x - 3y + 2 = 0$ and $6x - py + 7 = 0$ are perpendicular. Find p . Then find the equation of the line through $(-2, -1)$ parallel to $6x - py + 7 = 0$. **ICSE 2020, 3 marks**

SOLUTION

Slopes: $\frac{5}{3}$ and $\frac{6}{p}$. Perpendicular: $\frac{5}{3} \times \frac{6}{p} = -1$, so $p = -10$.

The second line is $6x + 10y + 7 = 0$, with slope $-\frac{3}{5}$.

Parallel line through $(-2, -1)$: $y + 1 = -\frac{3}{5}(x + 2)$, so $5y + 5 = -3x - 6$.

Answer: $p = -10$; the line is $3x + 5y + 11 = 0$

3 Inclination and slope

7. The points $A(x, y)$, $B(3, -2)$ and $C(4, -5)$ are collinear. Write y in terms of x . (a) $3x - 11$ (b) $11 - 3x$ (c) $3x - 7$ (d) $7 - 3x$ **ICSE 2024, 1 mark**

SOLUTION

Slope of $BC = \frac{-5 + 2}{4 - 3} = -3$. Collinear points give the same slope for AB .

$\frac{y + 2}{x - 3} = -3$, so $y = -3x + 9 - 2 = 7 - 3x$

Answer: (d)

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